

Drawing Comparison & Discrepancy Report: W10-175

Project: Paisley Tower (P22-132)

Piece Mark: W10-175

Drawings Evaluated:

- **Macro Output Drawing:** W10-175_-_1.pdf (Raw 1-Page Macro Output)
 - **Ground Truth / Approved Drawing:** P22-132.W10-175.Rev 2.pdf (Engineer Approved 4-Page Set)
 - **Evaluation Baseline:** Engineer Step-by-Step Dimension Hierarchy Rules (7 Levels)
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1. Executive Summary & Root Cause

Tekla's raw automatic dimensioning macro captures every physical cutout edge, joint boundary, reveal line, and embedded anchor as an isolated geometric entity. Consequently:

1. Dimension lines are placed out of sequence without maintaining a consistent offset from the panel perimeter.
2. Continuous dimension chains (such as wall openings) are fractured into multiple disjointed lines.
3. Hardware text labels and leader lines collide, creating unreadable overlaps.
4. Essential multi-sheet fabrication views (cross-sections, rebar placement plans, and bending schedules) are missing entirely from the macro output.

In the ground truth drawing (Rev 2), the engineer manually corrected these discrepancies following a strict priority hierarchy and geometric consolidation rules.

2. Engineer's 7-Tier Dimension Hierarchy (Page 1 Elevation View)

The engineer's standard establishes a strict 7-tier dimension stacking order moving outward from the panel face toward the drawing border:

Tier / Category	Engineer Baseline Rule	Ground Truth Drawing (Rev 2)	Macro Generated Drawing (W10-175)	Discrepancy & Defect Description
Tier 1: Lifters	Placed closest to the panel face (innermost string).	P-205 LIFTERS chain: 977 2114 2433 2152 1690 positioned innermost above the panel.	Dedicated lifter chain is missing; anchor tags are intermingled with general hardware.	Order Inversion: Lifter chain must be placed at the innermost offset.
Tier 2: Splicers	Dedicated string placed immediately outside Tier 1 (Lifters).	SP15-457, SP25-1067, and SP30-1219 arranged on distinct, clean offset lines: 1580 288 288 2290 1310 1437 1665 220.	Couplers are merged indiscriminately with grout tubes and base opening strings.	Stacking Violation: Rebar couplers lack an isolated, dedicated level line.
Tier 3: Bracing Inserts	Dedicated erection insert tier.	Explicitly labeled BRACING INSERTS (P-300): 2360 5106 1900.	Labeled generically as HARDWARE and stacked at an incorrect vertical elevation.	Classification Error: Part P-300 misclassified and misplaced.
Tier 4: Embed Hardware	Grout tubes and primary wall embeds.	Clear strings for GT76-254, GT75-1219, GT75-686 with staggered leader lines.	Heavy bounding box collisions; labels and lines directly overlap one another.	Collision Defect: Text de-cluttering and leader line offsets are missing.
Tier 5: Other Hardware	Secondary embeds,	Clean sub-lines for CNDT (T)	Generates a phantom component tag	Data Integrity Defect: Phantom tag

Tier / Category	Engineer Baseline Rule	Ground Truth Drawing (Rev 2)	Macro Generated Drawing (W10-175)	Discrepancy & Defect Description
	conduit, electrical boxes.	and EB-S-PL (B).	BCMBD-1-K (not present in the BOM).	generation due to missing BOM validation.
Tier 6: Openings	Clean, continuous panel cutout dimensions.	Single unified chain across the panel: 373 975 4928 975 2115 labeled PROFILE OPENING.	Fractured into two disjointed lines: Line 1 (373 975 8018) and Line 2 (975 2115).	Segmentation Error: Multiple window coordinates split across separate strings.
Tier 7: Overall	Total panel boundary (outermost string).	9366 PROFILE OVERALL sits at top and bottom outer perimeters with zero external clashes.	Hardware notes and recess dimension lines extend beyond the overall dimension string.	Boundary Violation: Overall panel dimension boundary is breached.

3. Mathematical Resolution of Profile Openings

3.1 Structural Reality

- **Total Panel Length:** 9366 mm
- **Window Openings:** Two identical window cutouts of 975 mm each
- **Left Solid Margin:** 373 mm
- **Right Solid Margin:** 2115 mm

3.2 Macro Error vs. Ground Truth Consolidation

- **Macro Line 1:** Labeled PROFILE OPENING, starts with 373, captures the first window 975, then projects a trailing residual span of 8018 mm to the panel end ($373 + 975 + 8018 = 9366 \text{ mm}$).
- **Macro Line 2:** Labeled PROFILE OPENING, placed below Line 1, capturing the second window 975 and right margin 2115.

- **Mathematical Consolidation:**
Subtracting the second window cutout and right margin from Line 1's trailing span:
$$8018 - (975 + 2115) = 4928 \text{ mm}$$
- **Resulting Continuous Chain:**
$$373 + 975 + 4928 + 975 + 2115 \quad (\Sigma = 9366 \text{ mm})$$

4. Multi-Sheet Drawing Set Comparison

Sheet	Ground Truth Drawing (Rev 2)	Macro Output Drawing (W10-175)	Status / Defect
Sheet 1	Top in Form Elevation, Formwork Edge Tags (END 1–5, SIDE A–D), Paint Boundary Zones, Hardware BOM.	Top in Form Elevation, 3D View, Incomplete Hardware BOM, Notes (Overcrowded).	All elements crammed onto a single page; paint zones and inner face tags missing.
Sheet 2	Section A-A (254 mm panel thickness, 152/102 mm ledge), Detail B-B (Drip Edge), Detail C-C (Sill), Detail D-D (V-Reveal), Clean 3D View.	Completely missing.	Critical section cuts, thickness dimensions, and reveal profiles are absent.
Sheet 3	Structural Reinforcement Layout: rebar placement, stirrups, tie spacings (5@300, 6@240, 4@203), bar marks (RB10M 0040, RB15M 0068, RB25M 0139).	Completely missing.	Rebar layout views and dimension chains are absent.
Sheet 4	Reinforcing Bill of Materials (BOM) table detailing quantities, lengths, bend types,	Completely missing.	Reinforcing schedule and shape diagrams are absent.

Sheet	Ground Truth Drawing (Rev 2)	Macro Output Drawing (W10-175)	Status / Defect
	and pin bending diagrams.		

5. Architectural Finishes, Edge Labels & General Notes

1. Architectural Paint & Reveals:

- **Ground Truth:** Specifies 4 paint finishes (*Benjamin Moore Thunder AF-685, Silhouette AF-655, Simply White OC-117, Etruscan AF 355*) and an explicit **NO PAINT** boundary zone (1857 \ | 229 \ | 3065 \ | 229 \ | 3745 \ | 241). Architectural reveals are grouped on a dedicated **REVEAL (B)** string.
- **Macro Output:** Completely omits paint finishes and boundary zones. Architectural reveals are conflated with generic **PROFILE RECESS** strings.

2. Formwork / Edge Identification:

- **Ground Truth:** Clear mold edge markings: **END 1** through **END 5**, and **SIDE A** through **SIDE D**.
- **Macro Output:** Interior window cutout faces are unmarked.

3. General Notes Block:

- **Ground Truth:** Fully populated with imperial design parameters: **28 DAY: 6000 PSI, STRIPPING: 3500 PSI, WEIGHT: 24822 LBS, CONC. VOLUME: 6.05 yd³, CHAMFER: 25 mm, FINISH: Trowel Side Screed / Form Side Paint.**
- **Macro Output:** Text labels are truncated and numerical values are missing (**28 DAY (MPa), STRIPPING (N)** rendered blank).